Answer **all** questions.

**1** A DNA library is a collection of DNA fragments that have been cloned into vectors so that researchers can identify and isolate the DNA fragments that interest them for further study. There are basically two kinds of libraries: genomic DNA and cDNA libraries.

**(a)** Outline how a cDNA library is produced.

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**(b)** Explain why the cDNA library is preferred over the genomic library by the researchers.

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The presence and absence of the restriction site generates a restriction fragment length polymorphism (RFLP). RFLP can be a useful biological tool, for example as a marker in chromosomal mapping.

**(c)** State **one** other ways in which RFLP can be used as a biological tool.

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**(d)** Outline how the relative positions between the particular gene and the two RFLP markers can be determined.

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RFLP analysis is often applied to genomic DNA after digesting with restriction enzymes and separating the DNA by gel electrophoresis. Gel electrophoresis of this DNA produces a smear of DNA, rather than distinct bands.

For a smear of digested DNA, additional steps involving DNA hybridisation are needed to analyse the RFLP pattern for a particular gene.

**(e)** Outline the process of DNA hybridisation that allows the RFLP pattern for a particular gene to be visualised.

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[Total: 14]

**2** Telomere length regulates the life span of a cell, in which the numbers of cell cycles are limited before the cell undergoes cell senescence. Thus, telomeres limit the capacity of a cell to divide. Fig. 2.1 shows the relationship between telomere length and the number of cell cycles for the different types of cells in the body.

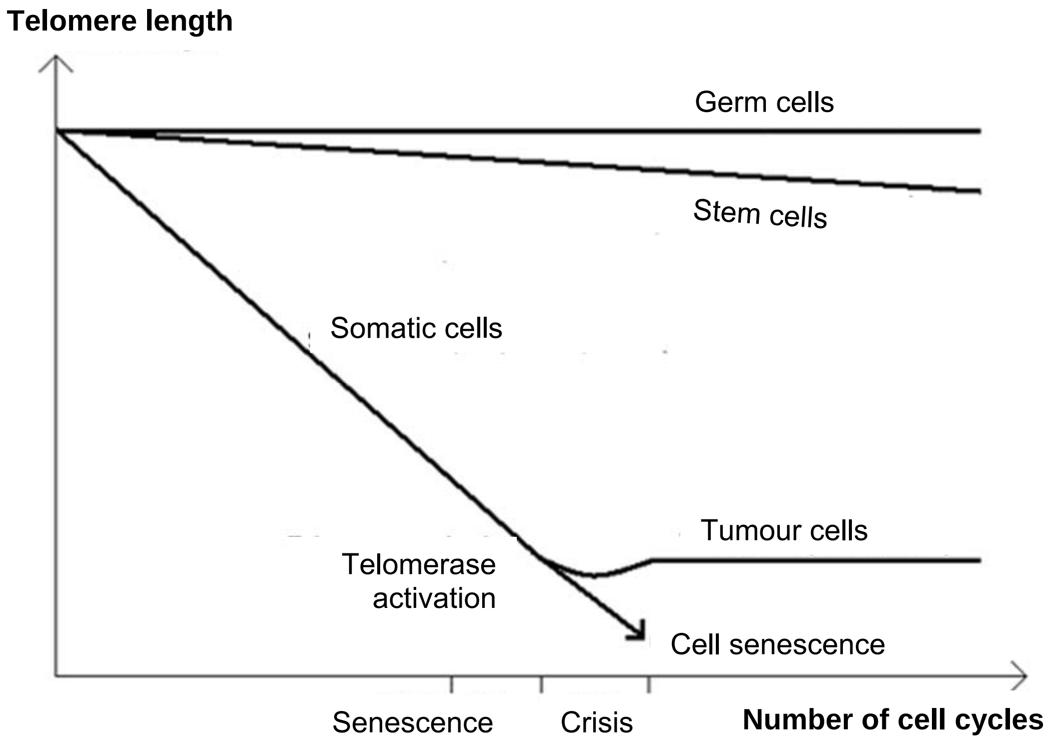


**Fig. 2.1**

With reference to Fig. 2.1,

**(a)** state one similarity in the relationship between telomere length and the number of cell cycles in stem cells and tumour cells.

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**(b)** explain how the relationship between telomere length and the number of cell cycles support one of the properties of stem cells.

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Cytokines bind Type I/II cytokine receptors that have a common receptor subunit called the common gamma-C chain (γc). Mutations of *γc* gene underlie X-linked severe combined immunodeficiency (X-SCID) and account for roughly half of all known cases of SCID. Fig 2.2 shows that the deficiency of γc protein blocks signalling by IL-7 and IL-15 which is required for haematopoietic stem cells to develop into natural killer (NK) cells and T cells. NK cells and T cells are lymphocytes that play critical roles in the immune system.

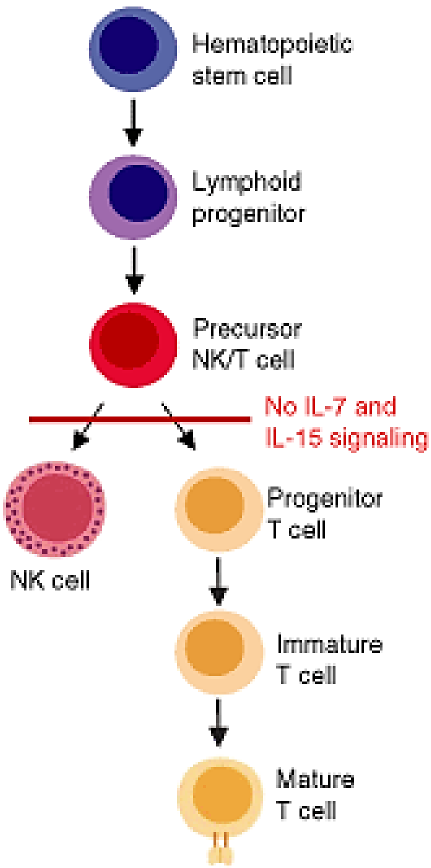


**Fig. 2.2**

**(c)** With reference to the information given and Fig. 2.2, explain why individuals with X-linked SCID have high mortality.

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Gene therapy has been used to treat patients with X-linked SCID. X-linked SCID is a recessive disease. Gene therapy uses a vector to deliver a normal copy of a γc gene to the cells where it is needed.

(d) Describe **one** non-viral gene delivery system that could have been used to insert normal copy of a γc gene into stem cells.

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**(e)** Explain why it is sufficient to insert a single normal copy of a γc gene into stem cells for gene therapy.

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[Total: 11]

**3** The *Aedes aegypti* mosquito is the main vector that transmits the viruses that cause dengue. Brazil is one of the first few countries to attempt to reduce the numbers of *A. aegypti* by using genetically modified (GM) male mosquitoes. This project was initiated a year after an epidemic of dengue fever that caused more than 1.5 million cases in Brazil.

The male *A. aegypti* mosquitoes are genetically engineered to contain a *t*TA gene that produces a protein called *t*TA. When they are released into the environment and mated with the female mosquitoes, the *t*TA gene is passed to the offspring, causing death. This will bring down the population numbers.

A heritable, fluorescent marker (GFP) gene was also introduced into the engineered mosquitoes to distinguish them from native pest insects and to help scientists with the management of pest control programmes.

**(a)** Explain why, in many examples of gene technology, fluorescent markers are used in preference to antibiotic resistance genes.

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Fig. 3.1 shows the transgene that has been added to the GM mosquito.

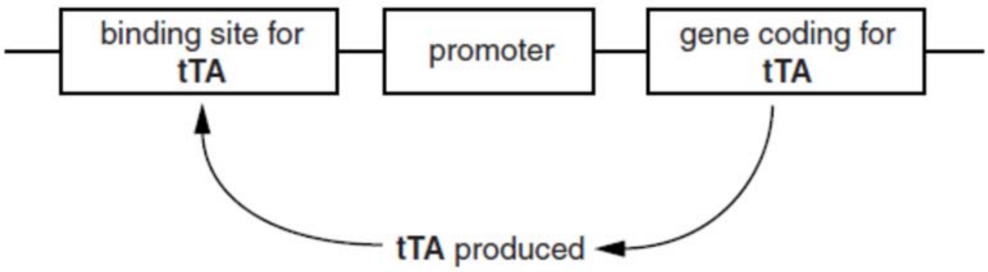


**Fig. 3.1**

**(b)** With reference to Fig. 3.1, explain why a promoter needs to be transferred along with the tTA gene.

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In addition to GM mosquitoes, there are experiments using mosquitoes infected with a bacterium, *Wolbachia that* seems to prevent them from spreading diseases.

The National Environmental Agency (NEA) of Singapore is conducting studies on the feasibility of using bacteria, *Wolbachia*. The bacterium is found naturally and used in the laboratory to infect male *A. aegypti* mosquitoes.

*Wolbachia*-infected male *A. aegypti* mosquitoes are released to mate with the females and their eggs do not hatch. This helps to suppress the Aedes mosquito population in Singapore.

**(c)** Describe reasons why some members of the public will prefer to use mosquitoes infected *Wolbachia,* instead of using genetically modified mosquitoes to decrease the population of Aedes mosquito.

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**(d)** The two methods of controlling mosquito population have some disadvantages. Suggest one disadvantage.

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Genetic engineering can also be used to make transgenic plants that increase the yield of cash crops. In 1907, scientists found that a common plant tumour was caused by the invasion of a bacterium called *Agrobacterium tumefaciens*. They discovered that the tumour was caused when plasmids from the bacteria were taken into the DNA of the host plant's cells and expressed. By 1983, biotechnologists were able to introduce genes into plant cells using modified plasmids that do not result in tumour formation. Fig. 3.2 shows the procedure of transforming *Brassica* using this method.

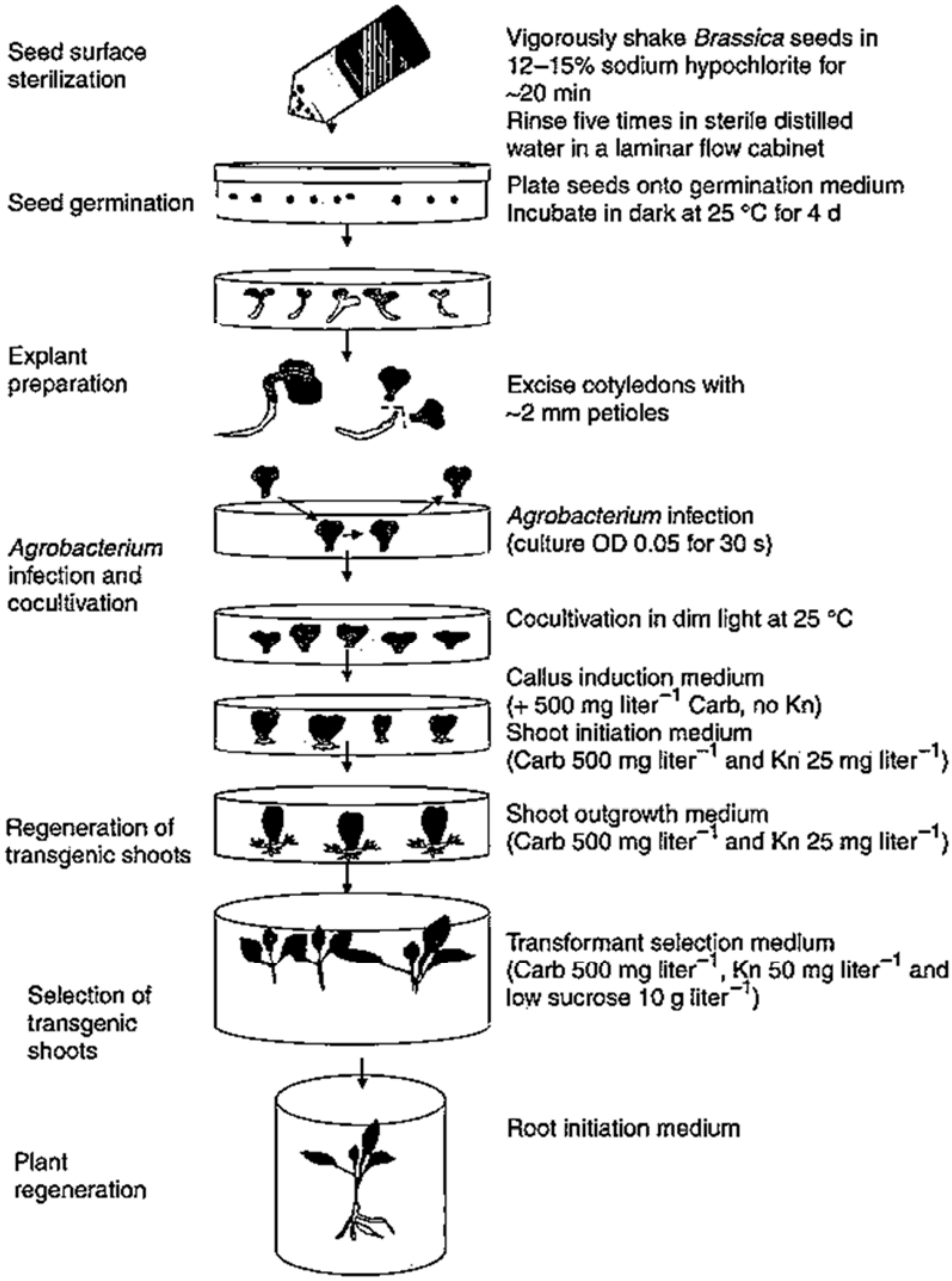


**Fig. 3.2**

**(e)** With reference to Fig. 3.2, explain why carbenicillin and kanamycin were added in the various media.

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**(f)** Explain why plant tissue culture technique is used to regenerate the transformed plants.

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**(g)** State the difference in composition between the root initiation medium and the shoot outgrowth medium.

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**(h)** Explain the significance of genetic engineering in improving the quality and yield of crop plants and animals using **named examples**.

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[Total: 15]

**4 Planning Question**

Enzymes such as catalase are protein molecules which are found in living cells. They are used to speed up specific reactions in the cells. They are all very specific as each enzyme just performs one particular reaction.

Catalase is an enzyme found in food such as potato and liver. It is used for removing hydrogen peroxide ($H_2O_2$) from the cells. Hydrogen Peroxide is the poisonous by-product of metabolism. Catalase speeds up the decomposition of Hydrogen Peroxide into water and oxygen.

Your planning is to investigate how the concentration of the enzyme affects the rate of reaction of the enzyme catalase on the decomposition of hydrogen peroxide. Rate of oxygen can be measured using a gas pressure sensor that is connected to an interface that translates sensor data. It is possible to measure the pressure of oxygen gas formed as $H_2O_2$ is destroyed. At the start of the reaction, there is no product, and the pressure is the same as the atmospheric pressure, being 101kPa.

Your planning must be based on the assumption that you have been provided with the following apparatus and materials, as well as other appropriate apparatus and equipment found in the school laboratory.

- 1 potato
- 1 electric blender
- cheese cloth
- Filter funnel
- 3% hydrogen peroxide
- pH Buffers
- Glass rods
- Knife
- Test tubes
- Stopwatch
- Access to water bath
- Gas Pressure sensor
- Interface and software
- Tubings with valves
- Other measuring instruments (measuring cylinder, syringes, etc.) and apparatus (beakers, test-tube, etc.)

Your plan should have a clear and helpful structure to include:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it, including scientific reasoning behind procedure.
- Be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- Include the layout of result tables and graphs with clear headings and labels.
- Include full details and explanations of the procedures that you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible.
- Include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

**5 Free response question**

Write your answers to this question on the separate answer paper provided.

Your answers:
- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set in sections **(a)**, **(b)** etc., as indicated in the question.

**(a)** Explain the problems associated with the expression of eukaryotic genes in prokaryotes and how these problems are overcome. [6]

**(b)** Explain the steps in gel electrophoresis. [7]

**(c)** Discuss the goals and benefits of the Human Genome Project. [7]

[Total: 20]